Your 6 Month Journey To Success



Your **6 months** journey

Start your journey with basics

Basics of Web Development

Getting started with Basics
Introduction to HTML
More on HTML
Forms
Intro to CSS
Styling with CSS
Starting with the Resume project
Flex
Completing Resume
Responsive designs
Animations and 3D space
Bootstrap

Getting started with JavaScript
Fundamentals of JS
Loops, Arrays, Function & Objects
Working of JS
Functions in JS
OOP in JS
ES6 Classes
DOM Manipulation
Mini-Project: Quiz Game
JS ASYNCHRONOUS
Modules in js
jQuery
Git

Front End with React

- Introduction to React
- Learning JSX
- Score-Keeper App
- Create-React-App
- React Components - I
- React Components - II
- Styling in React
- Component Lifecycle Methods
- React Hooks
- Firebase 9
- React Router v6
- Context API
- Redux in React
- Redux Toolkit

Move forward with either FRONTEND or BACKEND

Back End with Node.js

Move forward with either FRONTEND or BACKEND

Back End with Node.js

- Getting Started With NodeJS
- Modules in NodeJS
- Exploring More Modules
- Getting Started with Express
- Model-View-Controller - I
- Model-View-Controller - II
- File Upload, Session and Cookies
- Getting started with API
- Securing APIs
- Adding More features to E-Com App
- Handling Errors
- Getting Started with Databases
- MongoDB with NodeJS - I
- MongoDB with NodeJS - II
- Working with Mongoose - I
- Working with Mongoose - II
- Socket Programming
- Task Runners in NodeJS
- Deployment

Finish it basics of Prog. and DSA

Data Structures & Algorithms with Java/Python

- Basics of Programming
- Loops and Functions
- Arrays/Lists
- Strings and 2D Lists

Note: You can choose either **frontend (FE → React → DSA)** or **backend (FE → Node → DSA)** for your 6-month journey, and complete the other module in the next phase.

Curriculum for your success

Advanced modules to accelerate your career

Advanced Data Structures & Algorithms

- Problem Solving Techniques
- Object Oriented Programming
- Linear Data Structures
- Trees
- Advanced Data Structures
- Dynamic Programming

Operating Systems

- Introduction to OS
- Process Management
- Memory Management
- Concurrency
- Storage Management
- Case Study: Linux OS

System Design

System Design

- Architectural patterns
- Application characteristics I
- Application characteristics II
- Scaling Techniques
- Database
- Communication
- Web Applications
- Servers and security

Database Management System (DBMS)

- Introduction to DBMS
- Data Modeling
- Entity-Relationship Model
- Relational Model
- SQL
- Normalisation
- Transactions
- Classification of Databases
- NoSQL Databases
- Database optimisation

Aptitude Rounds

Aptitude Rounds

- Numbers
- Averages and Mixtures
- Arithmetic and Word Problems
- Counting
- Time, Speed and Distance
- Reasoning
- English
- Data Interpretation
- Miscellaneous Topics

Curriculum for your success

Elective modules to make you future ready

Generative AI

Build real-world applications using Generative AI tools

- Foundations of Generative AI & Prompt Engineering
- Hands-on with ChatGPT, Gemini & AI coding assistants
- Secure Authentication System using AI (JWT, MongoDB, SDLC)
- Full-Stack Social Media App using AI (React, Node.js)
- Code optimization, testing & deployment with GitHub Copilot
- LangChain fundamentals & text summarization
- AI Interview Simulator (resume parsing, question generation)
- End-to-end project deployment (Netlify & Render)

DevOps Essentials

DevOps Essentials

Automate, deploy, and scale modern applications

- DevOps culture, principles & best practices
- CI/CD pipelines for automated build, test & deployment
- Docker & containerized application deployment
- Kubernetes for container orchestration
- Infrastructure as Code using Terraform

Progressive Web App (PWA) Development

Build fast, reliable, and offline-ready web apps

- PWA fundamentals & architecture
- Service Workers & offline capabilities
- Lazy loading & code splitting for performance
- Caching strategies & cache versioning
- Testing & optimization of PWAs

Web3 Foundations

Web3 Foundations

Understand blockchain and decentralized technologies

- Web3 concepts & ecosystem overview
- Blockchain fundamentals & consensus mechanisms
- Smart contracts using Solidity
- Building & deploying decentralized applications (DApps)
- Cryptocurrency fundamentals & real-world use cases

Tools & Technologies

Tools covered in Tech Electives

and more.

Projects

Frontend

Dashboard Project

A personalized dashboard for a web-based platform that allows users to track their activities, view meetings and access various features. The dashboard should provide an engaging and visually appealing interface.

Responsive Flipkart Clone

Flipkart Home Page Clone, with the the addition of responsiveness by incorporating relative units and media queries in the project.

SASS Project

"Flipkart Clone" using SASS and SCSS to enhance the styling and maintainability of the project's CSS. The project replicate the appearance and layout of the Flipkart website's header, product categories, banners, and footer.

Resume Project

A website that effectively displays personal information, skills, work experience, and contact details. A website with modern design with various sections including an About section, Skills display, Work Experience.

Music Player App

A Music Player application that enables users to play locally stored songs, as well as create and update.

Stock Market Analysis

A web Application to show all available stocks in the portfolio and also their data over different ranges of time.

Seat Booking App

It is a JavaScript-based project that enables users to select movies from a dropdown, choose available seats, and calculate the total price dynamically. Users can book seats, receive confirmation alerts, and reset their selections. The app ensures that occupied seats cannot be selected and updates the UI in real time.

Ed-Tech Website Capstone Project

A comprehensive, responsive website for an EdTech company specializing in programming education. The website effectively showcases the company's course offerings, communicate its value proposition, and facilitate potential student engagement.

& more

Projects

Projects

Frontend (React.js)



IPOD

The React.js iPod project builds an interactive click-wheel menu where users rotate to navigate, tap to select, and use Menu to go back, with sections like Settings, Games, Music, and Coverflow. The Music section includes submenus such as All Songs and Artists.



Photofolio

An Online Photo Album that allows users to upload, organise, and share their digital photos.



BustBuy - I

An E-Commerce that enables individuals to browse available items, add or remove them from their shopping cart, and complete the purchasing process.



BusyBuy - II

An E-commerce application with React, integrating Redux for state management and Firebase for user authentication, improving real-time user experience

Backend (Node.js)



Easily



Postaway

Backend (Node.js)

Easily

A job portal website that allows recruiters to post and manage job listings and provides a user-friendly platform for job seekers to find and apply for suitable roles.

Postaway

An innovative tech startup with a vision to create a social networking platform. This platform enable users to create posts with text, media and support user engagement with likes and comments.

Flight Booking Application

A comprehensive flight booking system in Node.js that enables users to search, book, manage, and cancel flights, while providing administrators with full control over flight inventory and user management.

StoreFleet

An instant product delivery service similar to Blinkit, Zepto, and Swiggy. StoreFleet aims to be a formidable competitor by offering instant delivery not only for food and groceries but also for various products such as smartphones and other categories.

& more

Open Elective Projects

Generative AI Projects

- **Generative AI Foundations Project**
  Build a strong foundation in Generative AI using tools like ChatGPT, Gemini, prompt engineering techniques, and AI coding assistants.

- **Authentication Project Using GenAI**
  Develop a secure, AI-assisted authentication system with encrypted passwords, MongoDB integration, and JWT-based security.

- **Social Media Project Using ChatGPT (Part I)**
  Create a full-stack social media web application using React and Node.js, covering SDLC, API development, and test data generation.

- **Social Media Project Using ChatGPT and GitHub Copilot (Part II)**
  Optimize and deploy the social media application using GitHub Copilot, caching techniques, API testing, and load testing, with deployment on Netlify and Render.

- **LangChain and Text Summary Application**
  Build AI-powered applications using LangChain, including a text summarization model integrated with ChatGPT.

- **Ninja-Interviewer Project**
  Develop an AI-based interview simulator that analyzes resumes and dynamically generates technical interview questions, with end-to-end deployment.

Open Elective Projects

Open Elective Projects



Progressive Web App (PWA) Projects

- **Progressive Web App Fundamentals**
  This project introduces the principles and benefits of Progressive Web Apps. Learners understand core PWA features, architecture, and how PWAs enhance performance, reliability, and user experience.

- **Service Workers and Offline Capabilities**
  This project focuses on implementing service workers to enable offline access and background processes. Learners explore the service worker lifecycle, caching strategies, and handling network requests efficiently.

- **Performance Optimization with Lazy Loading and Code Splitting**
  In this project, learners optimize web application performance using lazy loading and code-splitting techniques. The focus is on reducing load times, improving runtime efficiency, and following best practices for performance optimization.

- **PWA Testing and Caching Strategies**
  This project emphasizes testing and reliability of PWAs. Learners implement testing strategies for service workers, manage caching policies, handle cache versioning, and ensure consistent performance across environments.

Open Elective Projects

Open Elective Projects



Web3 & Blockchain Projects

- **Introduction to Web3**

  This module introduces the fundamentals of Web3 and decentralized technologies. Learners explore key Web3 components, understand how decentralized systems work, and study real-world Web3 applications.

- **Blockchain Fundamentals**

  This project covers the core concepts of blockchain technology, including blockchain architecture, consensus mechanisms, and practical use cases across industries.

- **Decentralized Application (DApp) Development**

  In this project, learners design, build, and deploy decentralized applications. The focus is on DApp architecture, Ethereum-based development, and interaction with smart contracts.

- **Smart Contract Development**

  This module teaches learners how to write, deploy, and manage smart contracts using Solidity. It covers contract lifecycle management and interaction with decentralized applications.

- **Cryptocurrency Fundamentals**

  This project explores the cryptocurrency ecosystem, including how cryptocurrency transactions work, wallet concepts, and the relationship between cryptocurrencies and blockchain technology.

Open Elective Projects

Open Elective Projects



DevOps Projects

- **DevOps Foundations and Practices**

  This module introduces the principles, culture, and practices of DevOps. Learners explore real-world DevOps workflows, understand the benefits of DevOps in organizations, and study successful implementations that bridge development and operations.

- **CI/CD Pipeline Implementation**

  This project focuses on designing and implementing continuous integration and continuous delivery pipelines. Learners automate build, test, and deployment workflows, integrate CI/CD tools with other services, and gain hands-on experience in streamlining software delivery.

- **Docker and Containerization**

  In this project, learners use Docker to containerize applications and manage deployments efficiently. It covers creating Docker images and containers, understanding containerization concepts, and deploying applications using DockerHub.

- **Kubernetes Deployment and Management**

  This project teaches learners how to manage containerized applications using Kubernetes. It includes setting up a Kubernetes cluster, understanding Kubernetes architecture, and deploying and managing scalable applications in a production-like environment.

- **Infrastructure as Code with Terraform**

  This module focuses on automating infrastructure provisioning using Terraform. Learners write and apply Terraform configurations, manage infrastructure changes, and implement Infrastructure as Code best practices for scalable deployments.

Coding Ninjas Developer Clubs

By the end of the program you'll become a **Ninja Master Builder!**

Full-Stack Proficiency

Skills
UI, UX, server-side logic, database interactions

What you'll be able to do:
Build a social media platform where users can create profiles, post updates, and connect with others, all while managing the backend infrastructure to handle user data securely.

MongoDB. Express JS React node

Frameworks and Libraries

Skills
Leverage pre-built components and modules to streamline development

What you'll be able to do:
Use React to build a responsive user interface for a web application, or use Node.js and Express.js to create a lightweight and scalable backend server.

React

Job-Ready Skills

Skills
Be well-equipped for a variety of job opportunities

What you'll be able to do:
You could pursue roles such as a frontend developer, backend developer or a full stack developer depending on your interests and career goals.



Security Implementation

Skills
Protect web applications from cyber threats

What you'll be able to do:
Use authentication and authorization mechanisms to ensure that only authorised users have access to sensitive information.



Continuous Learning

Skills
Embrace a mindset of continuous learning

What you'll be able to do:
You could learn new programming languages, frameworks, or tools through online courses, workshops, or community events to expand your skill set and enhance your career prospects.



Problem-Solving Skills

Skills
Troubleshoot issues and find solutions to complex problems

What you'll be able to do:
You might encounter a bug in your code that is causing a website to crash, requiring you to analyze the code, identify the issue, and implement a fix.



Version Control Mastery

Skills
Track code changes and collaborate with team members

What you'll be able to do:
Use Git to manage different versions of your codebase, allowing you to easily revert to previous versions if needed.



Performance Optimization

Skills
Reduce server response times

What you'll be able to do:
Use image compression to reduce the size of images on a website, making the site load faster and improving the user experience.



Agile Development Practices

Skills
Emphasize iterative development, collaboration, and flexibility

What you'll be able to do:
Use Scrum to break down a large project into smaller, manageable tasks called sprints, allowing you to deliver incremental updates to the project and gather feedback from stakeholders along the way.

